Unsupervised Domain Adaptation Semantic Segmentation of Remote Sensing Images With Mask Enhancement and Balanced Sampling

Xin Li, *Senior Member, IEEE*, Yuanbo Qiu, Jixiu Liao, Fan Meng, and Peng Ren, *Senior Member, IEEE*

***Abstract*— Unsupervised domain adaptation (UDA) aims to improve model performance in the target domain by leveraging labeled data from the source domain while not requiring labeled data in the target domain. It has been widely applied in cross-domain semantic segmentation of remote sensing images (RSIs). Despite some advancements in this area, challenges such as class confusion due to color and texture similarities, class imbalance due to significant scale variations and sample imbalance continue to impede progress in UDA for RSI segmentation. To address these challenges, we propose a novel self-supervised teacher–student network framework, including two innovative techniques: mask-enhanced class mix (MECM) and scale-based rare class sampling (SRCS). The MECM method applies a high proportion of masks to mixed images derived from both source-domain images and target-domain images, which encourages the model to infer the semantic information of masked areas from the surrounding context, enhancing cross-domain contextual semantic learning and improving the recognition accuracy of similar classes. Additionally, SRCS increases the sampling proportion of small-scale rare classes, mitigating the issue of class imbalance. Experiments show that our method outperforms existing UDA techniques in terms of PA, mF1, and mIoU, achieving state-of-the-art results on three public datasets. Notably, in the Potsdam IRRG to Vaihingen UDA scenario, our method's performance on the key metric, mIoU, even surpasses that of supervised training, demonstrating the superiority of our approach. Codes are available at https://github.com/Qiuyb-ai/UDA-With-ME-and-BS.**

***Index Terms*— Data augmentation, remote sensing images (RSIs), semantic segmentation, unsupervised domain adaptation (UDA).**

I. Introduction

With the number of remote sensing images (RSIs) increasing extensively and rapidly, deep learning semantic segmentation technologies have undergone significant advancement, especially in remote sensing surface monitoring [2]. Semantic segmentation has been a crucial technique in satellite image processing [3], aiming at pixel-level classification of various semantic targets in an image. It has proved to be highly effective in applications such as land-cover classification [4], [5], road extraction [6], building detection [7], and farmland segmentation [8] using RSIs.

Over the past decade, the performance of data-driven deep learning semantic segmentation methods has dramatically improved, making them the predominant approach in RSI segmentation [6], [9]. However, this enhancement considerably depends on the availability of large-scale manually annotated data, which is often costly to produce [10]. When semantic segmentation models are applied to new scenes with insufficiently labeled data, their performance may significantly degrade compared to the source domain [11], [12]. This issue is particularly pronounced in remote sensing, where satellite images exhibit substantial visual differences due to varying geographical locations, seasons, imaging platforms, and conditions. Consequently, deep learning-based semantic segmentation models encounter difficulties in their generalization to unseen domains [13], [14].

Recently, unsupervised domain adaptation (UDA) has been proposed to address the problem of cross-domain semantic segmentation. This approach leverages knowledge from the source domain to enhance model performance in the target domain without requiring labeled data there [15], [16], [17]. In computer vision, numerous UDA methods have been proposed for natural image classification and semantic segmentation tasks. These methods primarily focus on minimizing distribution discrepancies between the source and target domains in the feature or output space. Current UDA approaches can be categorized into generative training methods [18], [19], [20], adversarial training methods [21], [22], and self-training (ST) methods [14], [23], [24]. Generative training methods employ image translation to convert source-domain images into the styles of the target domain at the pixel level, aiming to minimize the differences in the output space between domains [25]. Adversarial training methods use generative adversarial networks (GANs) to train a discriminator to distinguish whether the generator's output comes from the source domain or the target domain. Simultaneously, the generator is trained to make its output indistinguishable from the discriminator, aiming to minimize discrepancies

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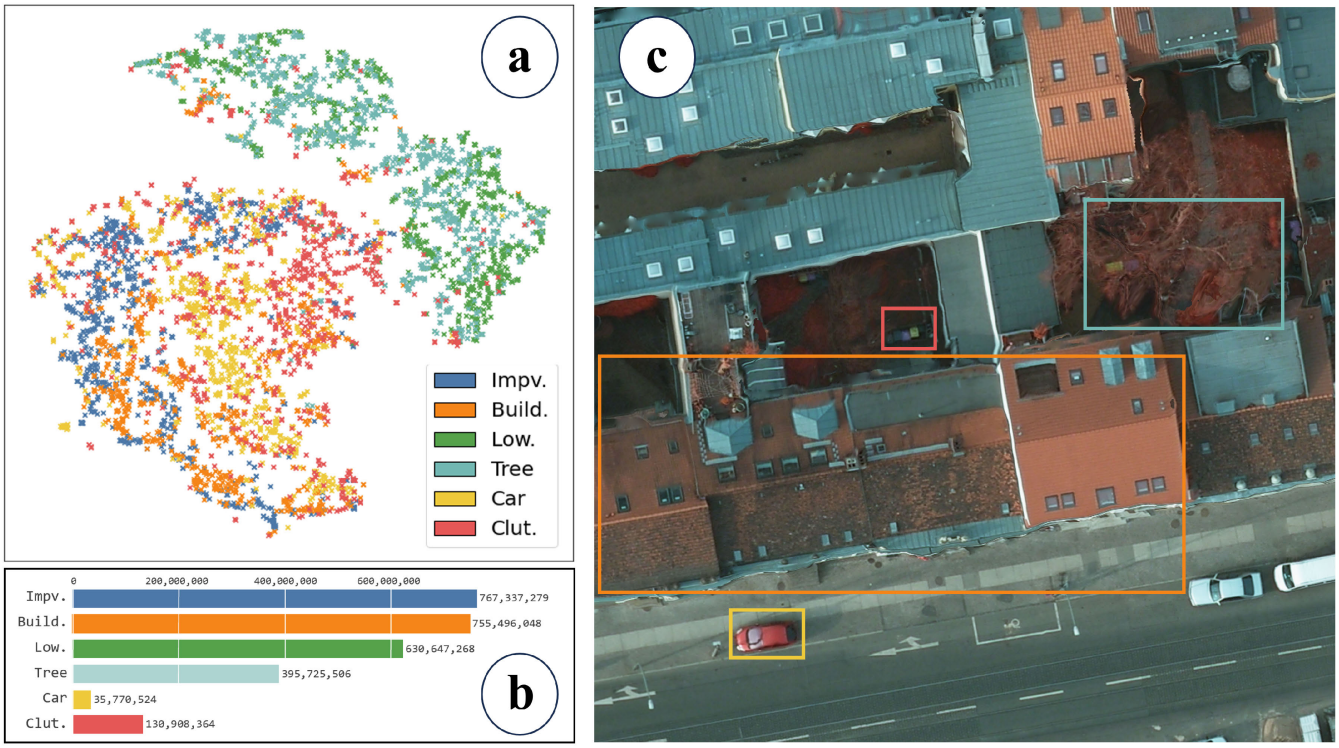


Fig. 1. Three challenges encountered in RSI segmentation. (a) t-SNE visualization of features extracted by the pretrained MiT-b4 network [1] on the Potsdam IR-R-G dataset demonstrates the existence of class confusion (C1), for example, the features of "Tree" and "Low Vegetation" classes are closely aligned, while the distribution of the "Clutter" class is highly irregular. (b) Statistics of the total pixel counts for each category in the Potsdam dataset reveal a significant class sample imbalance (C2a) issue. (c) Image from the Potsdam dataset illustrates that there are often significant scale differences (C2b) among different classes in RSIs.

in the feature space between domains [26]. However, both generative and adversarial training methods often suffer from instability and optimization difficulties. ST methods employ a teacher–student network framework, where the teacher network provides pseudo-labels for the target domain to guide the student network's learning. Compared to the first two methods, ST methods have been proved to be more effective in utilizing target-domain information and have shown excellent performance in cross-domain semantic segmentation of RSIs [27].

Although the recent UDA methods have achieved remarkable performance in cross-domain RSI segmentation, these methods still face the following challenges.

C1 (Class Confusion): Certain classes in RSIs, such as "Low Vegetation" and "Tree," share similar color and texture features, which leads to confusion in semantic segmentation models, as shown in Fig. 1(a). Additionally, RSIs often contain complex background information with heterogeneous feature distributions, complicating the learning process for these models.

C2 (Class Imbalance): Including sample imbalance and scale variations. a) Datasets often suffer from sample imbalance, with a large disparity in the total number of pixels across different classes. Fig. 1(b) shows that the pixel counts of the "Car" and "Clutter" classes are much lower than those of other classes. b) Moreover, in a single RSI, there is a significant scale variation among different classes. For instance, "Building" and "Car" classes exhibit substantial scale differences in Fig. 1(c). These imbalances can result in poor segmentation performance for small-scale and few-shot classes.

To address these challenges, we propose a novel self-supervised teacher–student network framework for mask-enhanced and balanced-sampling UDA. This framework includes two innovative techniques: mask-enhanced class mix (MECM) and scale-based rare class sampling (SRCS).

First, we propose the MECM method to address the challenge of class confusion (C1), which uses ClassMix to combine source- and target-domain images, creating a mixed image with corresponding mixed labels. We then apply a large patch mask to the mixed image, compelling the model to infer the classes of the masked areas using contextual information. This approach enhances cross-domain semantic learning and mitigates class confusion caused by similarities in color and texture among certain classes.

Second, to address the challenge of class imbalance (C2), we propose SRCS, a method that increases the sampling probability of few-shot and small-scale classes. SRCS replaces random sampling during training, greatly alleviating the issue of sampling imbalance (C2a), by calculating the total pixel count of each class in the source-domain dataset and then increasing the sampling probability for images containing rare classes. To address the issue of scale variations (C2b), during the process of sampling images for a specific class, we discard images with very few pixels of that class, thus increasing the overall sampling probability for small-scale samples. Additionally, we exclude images where any class

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occupies a significant portion of the pixels, thereby reducing the learning bias toward large-scale objects.

Our framework effectively addresses the challenges of RSIs, enhancing the robustness and generalization ability of cross-domain semantic segmentation. Our contributions can be summarized as follows.

1) To address the challenges of cross-domain semantic segmentation in RSIs, we propose a concise and novel UDA framework based on self-supervised teacher–student networks, incorporating two innovative methods: MECM and SRCS. This framework effectively mitigates issues of class confusion and class imbalance.
2) We propose the MECM method, which employs a high ratio of patch masking on combined images from both source and target domains. This approach aims to enhance the semantic segmentation model's ability to learn context, thereby alleviating class confusion.
3) The SRCS method improves the learning of rare classes by increasing the sampling probability of small-scale and few-shot classes in the source-domain dataset, thereby addressing class imbalance.
4) Our proposed framework achieves state-of-the-art performance on three public datasets. The performance on the key metric, mIoU, even surpasses that of supervised training in the Potsdam IRRG to Vaihingen UDA scenario, demonstrating the superiority of our approach.

II. Related Works

A. RSI Semantic Segmentation

Semantic segmentation has always been an essential technique in RSI processing applications, aiming to classify different semantic targets at the pixel level in a single image. This technology can be applied in various intelligent RSI applications, such as land-cover classification, road extraction, building extraction, and farmland classification. Since Long et al. [28] introduced convolutional neural networks (CNNs) for semantic segmentation, and deep learning models based on the CNN have been widely adopted in methods like U-Net [29], PSPNet [30] and DeeplabV3+ [31].

In recent years, Transformers [32] have achieved significant success in natural language processing and have been applied to various computer vision tasks, including image classification [33] and semantic segmentation. Zheng et al. [34] first proposed the Transformer-based semantic segmentation model SETR. However, SETR can only produce single-scale feature maps, limiting its ability to effectively capture contextual information, which is crucial for semantic segmentation tasks. Hong et al. [35] proposed a novel 3D generative pretrained Transformer, SpectralGPT, for spectral RSI segmentation. Xie et al. [1] introduced Segformer, a series of MixTransformer backbone networks specifically designed for semantic segmentation to extract multiscale features, achieving state-of-the-art performance in terms of efficiency, accuracy, and robustness.

Recent studies show that Transformer-based methods surpass CNN-based methods in semantic segmentation UDA [36]. However, many CNN-based methods, such as PFST [14], still dominate the RSI semantic segmentation field, using models like DeeplabV3+ [31] as decoders and ResNet-50 [37] as encoders. Since CNN-based methods like DeeplabV3+ are less effective than the latest Transformer-based methods, we choose to use the Transformer-based Segformer in our UDA framework. Our experiments demonstrate that this significantly improves UDA performance.

B. RSI UDA Semantic Segmentation

UDA semantic segmentation (UDASS) aims to adapt a semantic segmentation model trained on a labeled source domain to an unlabeled target domain. This technique addresses significant visual discrepancies between satellite images obtained from different geographic locations and imaging conditions, enhancing the model's robustness and generalization across various data distributions. These UDA methods are categorized into generative training methods, adversarial training methods, and ST methods.

Generative training methods use generators to learn the feature distributions of both the source and target domains, narrowing the gap between them. Typically, these methods employ GANs [38], where the generator creates source-domain data that matches the target-domain features, and the discriminator differentiates between real target-domain data and generated data. This process helps the model learn the target domain's feature distribution, improving segmentation performance. For instance, Hoffman et al. [25] introduced CyCADA, which uses cycle consistency loss to convert source-domain images into target-domain style while maintaining semantic consistency, achieving effective domain adaptation. Li et al. [19] developed a semantic-preserved GAN, leveraging translated and unlabeled target images with a class distribution alignment module to train a domain adaptive semantic segmentation model. Zhao et al. [18] proposed resize-residual DualGAN, incorporating an in-network resizer module to address the scale discrepancies in RS datasets and a residual connection to enhance the stability of real-to-real image translation.

Adversarial training methods reduce distribution differences between the source and target domains through adversarial learning. In these methods, the generator and the discriminator in an adversarial network undergo adversarial training, making it challenging to distinguish between source-domain data generated by the generator and target-domain data, thereby improving performance on the target domain. Hoffman was the first to apply adversarial learning to align the feature maps extracted by the semantic segmentation network across different domains on a global scale [39]. Tsai et al. [26] proposed AdaptSegNet, which aligns the feature spaces of the source and target domains via adversarial learning, enhancing semantic segmentation accuracy. ADVENT [40] introduced entropy adversarial loss to enforce high prediction certainty for target predictions, achieving domain adaptation.

ST methods adapt the model by generating pseudo-labels for target-domain data and using these labels for training. Typically, this involves two steps: first, the model trained on the source domain predicts the target-domain data to generate

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pseudo-labels; second, the target-domain data with pseudo-labels is used as the training set to further optimize the model. For example, Zou et al. [41] improved ST effectiveness by adaptively selecting high-confidence pseudo-labels and employing a class-balanced strategy. Hong et al. [42] proposed dc-Net, which employs a self-supervised learning module to achieve hyperspectral image super-resolution. Tranheden et al. [43] proposed DACS, training the model on both labeled data and mixed data of images while corresponding labels and pseudo-labels from both domains to mitigate the low-quality pseudo-labels caused by domain transfer. Hoyer et al. [44] introduced the masked image consistency strategy, training the model on masked target images to enhance contextual learning.

Despite the advancements in UDA for RSI semantic segmentation, the performance of these UDA methods is still limited by challenges such as class confusion (C1) and class imbalance (C2). Our proposed methods aim to address these specific challenges, enhancing the performance and robustness of semantic segmentation models in cross-domain RSI scenarios.

C. Data Augmentation

In recent years, data augmentation has become crucial in UDA, especially with significant advancements in self-supervised learning. DACS [43] has proved the effectiveness of data mixing in UDA for semantic segmentation tasks. It uses ClassMix [45] to blend images from both the source and target domains, along with their labels and pseudo-labels, training on this mixed data to mitigate low-quality pseudo-labels caused by domain shifts. DAFormer [36] proposed a rare class sampling strategy (RCS) that statistically balances the distribution of semantic classes within a dataset, addressing the long-tail problem and enhancing the model's generalization ability. Yaras et al. [46] introduced randomized histogram matching, a fast real-time unsupervised training enhancement method that applies a simple color data augmentation strategy to RSIs, achieving superior results. CCDA [47] introduced a curriculum-style local-to-global cross-domain adaptation framework. This approach adapts from easy to hard based on adaptation difficulties determined by an entropy-based score for each patch of the target domain. MIC [44] leverages masked images in a novel way to learn context relationships for domain adaptation. During target-domain image training, it performs reconstruction to enhance the context-learning capability for UDA. However, this additional training process slows down the overall model training.

Although these data augmentation methods have achieved significant results in UDA, their effectiveness in RSIs remains uncertain. For example, ClassMix might hinder learning on target-domain data due to the excessive mixing of large-scale objects from the source domain, while RCS, which focuses solely on the overall distribution of classes in the dataset, struggles with the scale variations present in RSIs.

In summary, UDA methods have evolved to incorporate various strategies, including discrepancy minimization, adversarial training, ST, and data augmentation. Although these techniques have demonstrated promise in reducing domain discrepancies and enhancing model generalization, they still encounter significant challenges, such as class confusion and class imbalance. This proposes a self-supervised teacher–student network framework to address the challenges of RSIs, including two novel data augmentation methods. A detailed analysis of our framework will be provided in Section III.

III. Methods

This section provides a detailed description of the proposed UDA framework, with the overall architecture illustrated in Fig. 2. In Section I, we introduce the ST UDA framework that forms the basis of our method. Section II details the MECM method. Section III introduces the total loss function used for training in our method. Section IV provides detailed explanations of the SRCS method.

A. ST UDA Framework

In UDA, the neural network g_θ is trained using source-domain images $\boldsymbol{X}_S = \{\boldsymbol{x}_S^{(i)}\}_{i=1}^{N_S}$ and one-hot labels $\boldsymbol{Y}_S = \{\boldsymbol{y}_S^{(i)}\}_{i=1}^{N_S}$ in an attempt to perform well on the target domain $\boldsymbol{X}_T = \{\boldsymbol{x}_T^{(i)}\}_{i=1}^{N_T}$ without access to the target labels $\boldsymbol{Y}_T$. N_S and N_T indicate the sizes of the source datasets and the target datasets, respectively. The model trained only on the source domain is formulated as

$$\mathcal{L}_S^{(i)} = -\sum_{j=1}^{H\times W}\sum_{c=1}^{C} y_S^{(i,j,c)} \log g_\theta\left(\boldsymbol{x}_S^{(i)}\right)(j,c). \tag{1}$$

However, since the network cannot access information from the target domain, the model finds it difficult to generalize to the target domain. To address the domain gap, we use the ST framework, which has been proved to have better performance. The ST framework uses a teacher network h_ϕ to generate pseudo-labels for the target-domain data

$$p_T^{(i,j,c)} = \left[c = \arg\max_{c'} \; h_\phi\left(\boldsymbol{x}_T^{(i)}\right)(j,c')\right] \tag{2}$$

where $[\cdot]$ denotes the Iverson bracket. Gradients are not backpropagated to the teacher network. Additionally, a quality/confidence estimate for the pseudo-labels is generated. Here, we use the ratio of pixels exceeding the threshold τ of the maximum softmax probability

$$q_T^{(i)} = \frac{\sum_{j=1}^{H\times W}\left[\max_{c'} h_\phi\left(\boldsymbol{x}_T^{(i)}\right)(j,c') > \tau\right]}{H\cdot W}. \tag{3}$$

The pseudo-labels and their quality estimates are typically used to further train the network on the target domain

$$\mathcal{L}_T^{(i)} = -\sum_{j=1}^{H\times W}\sum_{c=1}^{C} q_T^{(i)} p_T^{(i,j,c)} \log g_\theta\left(\boldsymbol{x}_T^{(i)}\right)(j,c). \tag{4}$$

During training, h_ϕ is updated based on g_θ. After each training step t, the exponential moving average of g_θ's weights is used

$$\phi^{t+1} \leftarrow \alpha\phi^t + (1-\alpha)\theta^t. \tag{5}$$

In our work, we do not directly use the target-domain images $\boldsymbol{X}_T$ and the pseudo-labels $p_T^{(i,j,c)}$ generated by the teacher

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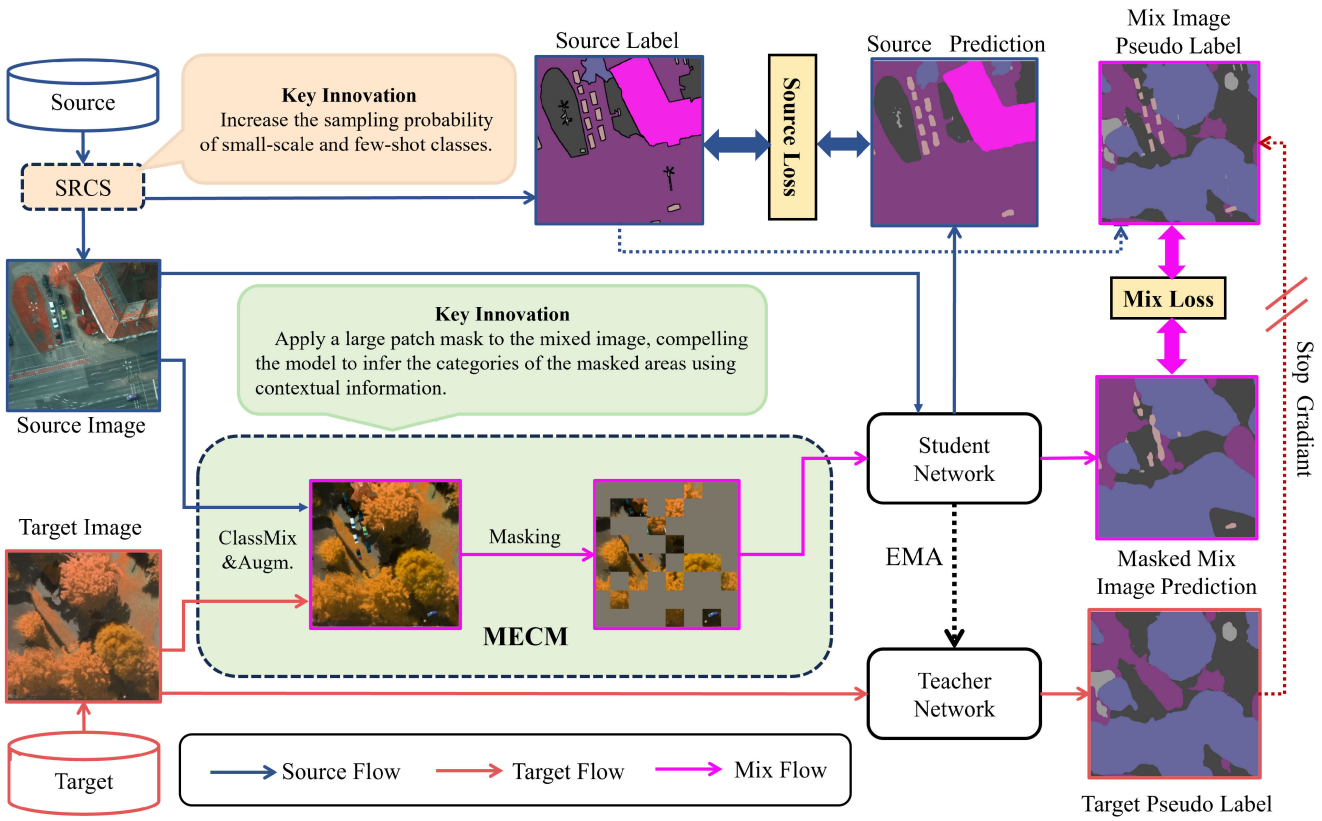


Fig. 2. Overview of our proposed self-supervised teacher–student network framework for UDA. We employ a self-learning teacher–student network, where the teacher model generates pseudo-labels from target images to supervise the student model. The student model is trained on both source images and mixed images created using ClassMix, which heavily masks portions of source images and target images to enhance contextual learning. The teacher model is updated using an Exponential Moving Average (EMA) approach. Furthermore, before feeding data into the model, we implement SRCS on the source dataset. This method increases the sampling probability of classes with small scales and few-shot samples to ensure balanced training across different classes.

model h_ϕ to train the student model g_θ. Instead, we employ the MECM method, which will be explained in detail in Section III-B. This method involves using high-proportion masks on class mixed images $\boldsymbol{X}_{\text{mix}}$ from both the source domain $\boldsymbol{X}_S$ and the target domain $\boldsymbol{X}_T$. The class mixed images $\boldsymbol{X}_{\text{mix}}$ are generated as follows:

$$\boldsymbol{X}_{\text{mix}}^{(i)} = M \odot \boldsymbol{X}_S^{(i)} + (1 - M) \odot \boldsymbol{X}_T^{(i)} \tag{6}$$

where M is a binary mask and $\odot$ denotes the element-wise multiplication. The corresponding mixed labels $\boldsymbol{Y}_{\text{mix}}$ are defined as

$$\boldsymbol{Y}_{\text{mix}}^{(i,j,c)} = M \odot \boldsymbol{Y}_S^{(i,j,c)} + (1 - M) \odot p_T^{(i,j,c)}. \tag{7}$$

The training is conducted on the masked mixed images $\boldsymbol{X}_{\text{mix}}$ and mixed labels $\boldsymbol{Y}_{\text{mix}}$, encouraging the model to infer the semantic information of the masked regions from the context. This enhances the model's ability to learn cross-domain contextual semantics.

B. Mask-Enhanced Class Mix

In previous unsupervised adaptation methods, ClassMix is often utilized to assist models in learning domain-invariant features across different domains. However, combining ClassMix source-domain images with target-domain images can result in several issues.

For example, in RSIs, a segmented single image may exhibit large-scale targets that occupy most of the space. The mixed image produced in such cases can hinder the network's ability to learn domain adaptation features. Moreover, due to the vast differences in target size, color, and texture within RSIs, the mixed image may display distinct boundaries, potentially biasing the network's learning process.

To address this, we employ a mask to shield ClassMix's training input. Specifically, we use ClassMix to generate mixed images from the source and target domains, denoted as $\boldsymbol{X}_{\text{mix}}^{(i)}$. The source-domain labels and the target-domain labels, generated by the teacher network, are combined to form $\boldsymbol{Y}_{\text{mix}}^{(i,j,c)}$. We then randomly sample patch masks M. Each mask is divided into $n \times n$ patches. For each patch, the mask M is randomly sampled from a uniform distribution

$$M_{i,j} = \begin{cases} 1, & \text{if } \epsilon_{i,j} > p \\ 0, & \text{otherwise} \end{cases} \quad \text{with } \epsilon_{i,j} \sim \mathcal{N}(0,1),\ 0 \le i,\ j < n \tag{8}$$

where p denotes the overall mask ratio. If the random number ϵ drawn from the standard normal distribution $\mathcal{N}(0,1)$ is greater than the specified probability p, the mask for that patch is set to 1, otherwise it is set to 0.

We then apply these masks to the mixed images $\boldsymbol{X}_{\text{mix}}^{(i)}$

$$\boldsymbol{X}_{\text{mix}}^{(i)'} = \boldsymbol{X}_{\text{mix}}^{(i)} \odot M. \tag{9}$$

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The semantic segmentation model, denoted as g_θ, is required to infer the complete semantic labels of each pixel from the masked mixed images and compute the loss with $\boldsymbol{Y}_{\text{mix}}^{(i,j,c)}$

$$\mathcal{L}_{\text{mix}}^{(i)} = -\sum_{j=1}^{H\times W}\sum_{c=1}^{C} \boldsymbol{Y}_{\text{mix}}^{(i,j,c)} \log g_\theta\left(\boldsymbol{X}_{\text{mix}}^{(i)'}\right)(j,c). \tag{10}$$

This approach compels the model to deduce the semantic labels of the obscured regions using contextual information. It mitigates the significant disruption caused by large-scale samples in RSIs during mixed image training and enhances the model's comprehension of spatial context.

C. Total Loss Function

The final training loss is composed of the source-domain loss $\mathcal{L}_S$ and the masked mixed loss $\mathcal{L}_{\text{mix}}$. The total loss is defined as

$$\mathcal{L} = \mathcal{L}_S + \mathcal{L}_{\text{mix}}. \tag{11}$$

This formulation ensures that the model leverages both the reliable labels from the source domain and the pseudo-labels from the target domain, while also benefiting from the contextual learning facilitated by the MECM method.

D. Scale-Based Rare Class Sampling

Although our proposed MECM can enhance the model's contextual learning, we have observed several factors that may affect the model's performance.

1) *Sample Imbalance:* An imbalanced dataset can prevent the model from effectively learning features of minority classes, leading to poor prediction performance for these classes.
2) *Dominance of a Single Class in the Source Domain:* If a particular class dominates the source domain, the model might ignore interclass contextual relationships. This dominance can also result in mixed images being largely covered by this single class, impairing the model's ability to learn useful features for cross-domain segmentation tasks.
3) *Coverage Problem:* The masking operation in ClassMix may entirely cover small-scale objects, reducing the model's ability to learn from these classes. This exacerbates the class imbalance problem. Experiments indicate that MECM decreases segmentation accuracy for small-scale classes such as "Car" and "Clutter."

To address these issues, we propose SRCS. When selecting images from the source-domain dataset, we weight the classes based on their pixel counts using a softmax function to ensure they are more frequently included in the training. This process can be formulated as follows:

$$P(c) = \frac{e^{(1-f_c)/T}}{\sum_{c'=1}^{C} e^{(1-f_{c'})/T}} \tag{12}$$

where $P(c)$ is the probability of selecting an image containing class c, f_c is the fraction of pixels of class c in the image, T is a temperature parameter that controls the smoothness of the distribution, and C is the total number of classes. We also set a threshold t to exclude images where a single class exceeds this pixel count, minimizing the impact of class dominance. This can be expressed as

$$I_t = \{I \,|\, \max(f_c) \le t\} \tag{13}$$

where I_t is the set of images selected for training and $\max(f_c)$ is the maximum fraction of any class in the image.

Furthermore, we establish a criterion to evaluate the pixel distribution in the source-domain dataset. We select only those images where the pixel count for a given class exceeds a certain threshold by excluding the top r images with the least pixel counts for that class. This ensures that the targeted classes have sufficient representation, mitigating the coverage problem of MECM. The selection criterion can be defined as

$$I_r = \{I \,|\, I \notin \text{Top}_r(c)\} \tag{14}$$

where $\text{Top}_r(c)$ represents the set of top r images with the least pixel count for class c.

Combining these criteria, the overall filtered image set can be summarized as

$$I_f = \{I \,|\, I \in I_t \wedge I \in I_r\}. \tag{15}$$

Finally, the SRCS process involves randomly selecting images from I_f based on the probability $P(c)$

$$I_{\text{SRCS}} = \text{RandomSelect}(I_f, P(c)). \tag{16}$$

Compared to randomly sampling images from the source domain, SRCS increases the probability of sampling images that contain small scales and few-shot classes. This helps to mitigate the class imbalance problem. Additionally, SRCS filters out images that have an excessive number of pixels belonging to a single category. This filtering is done to prevent large-scale source-domain images from overwhelming target-domain images in ClassMix, which could potentially degrade UDA performance.

IV. Experiments

A. Datasets and UDA Settings

To validate the effectiveness and robustness of our proposed method, we conduct experiments on publicly available datasets, specifically aerial images and their corresponding semantic segmentation labels from Vaihingen and Potsdam. The Vaihingen dataset comprises 33 images (2000 × 2000 pixels) with three spectral bands (near-infrared, red, and green). The Potsdam dataset includes 38 images (6000 × 6000 pixels) with four spectral bands (near-infrared, red, green, and blue). We further divide the Potsdam dataset into two subsets: Potsdam IRRG, containing the near-infrared, red, and green bands, and Potsdam RGB, containing the red, green, and blue bands. Following the PFST [14], we segment images from all three datasets into patches of 1024 × 1024 pixels with a stride of 512 pixels to increase the number of samples. These datasets are split into training and testing sets according to the official guidelines.

Using these three public datasets, we establish three different UDA settings to evaluate our method's performance. The details are as follows.

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TABLE I

EXPERIMENTAL RESULTS (%) ON THE POTSDAM IRRG TO VAIHINGEN SETTING. THE VALUES IN BOLD ARE THE BEST. METRICS WITH "*" INDICATE THAT THE METRIC SURPASSES THOSE OBTAINED BY SUPERVISED TRAINING METHODS ON THE TARGET DOMAIN

Method	IoU						PA	mF1	mIoU
	Imp. Sur.	Building	Low. Veg.	Tree	Car	Clutter			
Source-Only	72.16	84.57	47.25	71.96	55.69	18.94	81.13	71.12	58.43
RDG	77.37	87.68	58.78	73.90	66.40	20.28	84.90	75.54	64.07
DRDG	75.75	87.27	53.30	71.84	**70.02**	26.44	83.65	76.12	64.10
PFST	81.95	90.45	55.04	73.41	63.16	46.63	86.52	80.28	68.42
ST-DASegNet	80.90	86.20	46.78	67.76	50.05	32.65	83.22	74.09	61.10
DAFormer	82.62	88.65	54.42	74.12	57.52	**50.02**	86.41	79.97	67.89
MIC	82.14	88.83	60.54	77.87	62.88	42.61	87.61	80.70	69.15
Ours	**84.25**	**92.98**	**66.79**	**78.80**	69.43	41.43	**89.35**	**82.76***	**72.28***
Target-Only	86.44	91.71	71.60	81.00	75.01	23.92	90.58	80.95	71.62

1) *Potsdam IRRG to Vaihingen:* In this setting, the Potsdam IRRG dataset serves as the source domain, and the Vaihingen dataset serves as the target domain. We train the UDA model using the labeled training sets of the Potsdam IRRG dataset and the unlabeled training sets of the Vaihingen dataset, reporting the results on the Vaihingen testing sets.
2) *Potsdam RGB to Vaihingen:* In this setting, the Potsdam RGB dataset is the source domain, and the Vaihingen dataset is the target domain. The UDA model is trained on the labeled training sets of the Potsdam RGB dataset and the unlabeled training sets of the Vaihingen dataset, with results reported on the Vaihingen testing sets. To align the IR-R-G bands from Potsdam with the RGB channels in the model, we map each band as follows: near-infrared (IR) $\rightarrow$ red (R), red (R) $\rightarrow$ green (G), and green (G) $\rightarrow$ blue (B). This mapping enables consistent input structure across datasets, ensuring compatibility with the network input.
3) *Vaihingen to Potsdam IRRG:* In this setting, the Vaihingen dataset (IRRG) is the source domain, and the Potsdam IRRG dataset is the target domain. We train the UDA model on the labeled training sets of the Vaihingen dataset and the unlabeled training sets of the Potsdam IRRG dataset, with performance evaluated on the Potsdam IRRG testing sets.

B. Implementation Details

We re-implement the current state-of-the-art UDA methods for evaluation and comparison. These methods include RDG [18], DRDG [20], PFST [14], ST-DASegNet [23], DAFormer [36], and MIC [44]. We also consider evaluating the effectiveness of domain adaptation methods without using UDA techniques, termed "Source-Only." This approach involves training solely on the source-domain data and testing on the target domain. Due to the discrepancies between the source and target domains, the performance of this method significantly decreases. Additionally, we consider the "Target-Only" approach, where supervised training is performed directly on the target domain using both data and labels. A comparison of this result with that of the UDA methods shows the upper limitation in the performance of the latter.

To ensure a fair comparison, all methods employ the Segformer [1] segmentation network architecture, with MiT-b4 as the encoder and a decoder hidden size set to 256. The same data loading pipeline, optimizer, training process, and data augmentation are used based on the MMLab [48] training framework.

We apply the same data normalization and augmentation as PFST [14] for weak augmentation, including random resizing, cropping, random horizontal and vertical flipping, random rotation of 90°, 180°, or 270°, and random photometric distortion. For strong augmentation, we use the same color jitter and random blur as DACS [43] and replace ClassMix [45] with MECM. The AdamW [49] optimizer, with the parameters β_1 and β_2 set to 0.9 and 0.999, is used with a learning rate of 6×10^{-5} for the encoder and 6×10^{-4} for the decoder, over 40 k training iterations, with a batch size of 2, linear learning rate warm-up for 1500 iterations, and an EMA factor $\alpha = 0.999$. All experiments are conducted on a single NVIDIA GTX 2080Ti GPU with the PyTorch library.

Referring to [23] and [50], the pixel accuracy (PA), intersection over union (IoU) per class, and mean value of IoU (mIoU) and mean value of $F1$-score (mF1) is used to quantitatively evaluate the segmentation performance of the different methods.

C. Quantitative Results

We present the quantitative comparison results of different UDA methods under three distinct settings in Tables I–III.

1) *Potsdam IRRG to Vaihingen Setting:* Table I illustrates that models trained using UDA methods achieve substantial improvements in segmentation performance in the target domain compared to those trained solely on the source domain. Our proposed method not only outperforms the best current UDA methods in several metrics but also, in some cases, surpasses models that are trained with supervised learning on the target domain. Considering the Potsdam IRRG to Vaihingen setting, where the source-domain dataset contains significantly more images than the target-domain dataset, this demonstrates the potential of UDA methods to outperform models trained with supervised learning on smaller datasets, particularly when there is abundant source data. Moreover, in segmenting the challenging "Low Vegetation" and "Tree" classes, our method markedly outperforms others, indicating its superior ability to learn contextual features that distinguish similar classes.

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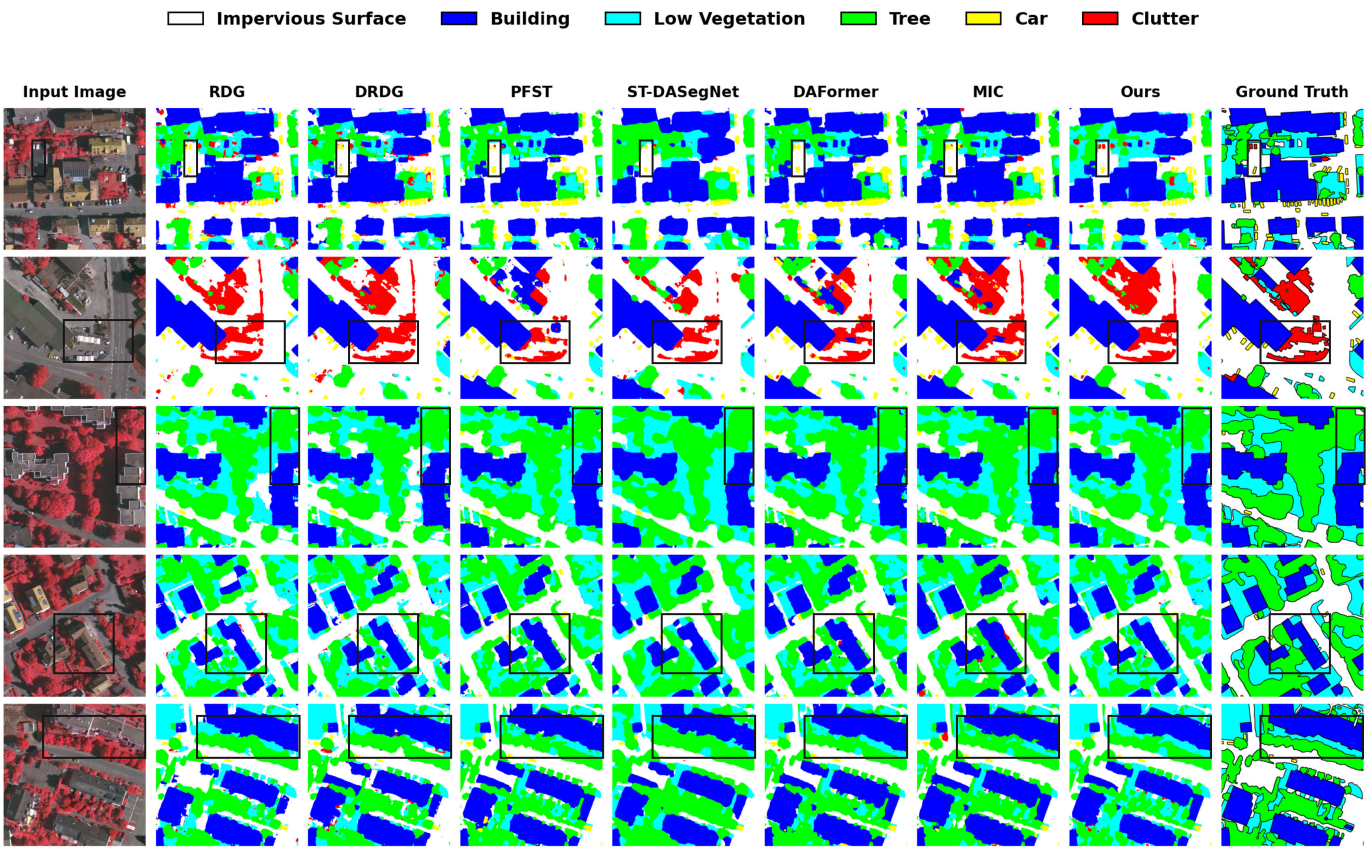


Fig. 3. Visualized semantic segmentation results on the Potsdam IRRG to Vaihingen setting.

TABLE II

EXPERIMENTAL RESULTS (%) ON THE POTSDAM RGB TO VAIHINGEN SETTING. THE VALUES IN BOLD ARE THE BEST. METRICS WITH "*" INDICATE THAT THE METRIC SURPASSES THOSE OBTAINED BY SUPERVISED TRAINING METHODS ON THE TARGET DOMAIN

Method	IoU						PA	mF1	mIoU
	Imp. Sur.	Building	Low. Veg.	Tree	Car	Clutter			
Source-Only	64.34	74.30	4.46	52.88	44.71	2.17	57.18	51.22	40.48
RDG	72.09	88.01	43.77	72.73	66.37	11.18	80.26	70.41	59.04
DRDG	65.57	85.36	42.46	68.56	**67.59**	13.32	78.15	69.41	57.14
PFST	79.31	88.56	50.43	70.99	63.31	32.15	84.32	76.45	64.13
ST-DASegNet	79.12	86.57	46.78	64.70	47.94	38.31	82.19	73.95	60.57
DAFormer	79.92	85.26	50.96	68.64	59.75	26.09	83.57	74.33	61.77
MIC	**82.56**	89.44	55.18	73.71	42.32	8.26	86.12	69.27	58.58
Ours	74.79	**92.44**	**62.82**	**78.74**	67.36	**42.47**	**86.80**	**81.17***	**69.77**
Target-Only	86.44	91.71	71.60	81.00	75.01	23.92	90.58	80.95	71.62

TABLE III

EXPERIMENTAL RESULTS (%) ON THE VAIHINGEN TO POTSDAM IRRG SETTING. THE VALUES IN BOLD ARE THE BEST

Method	IoU						PA	mF1	mIoU
	Imp. Sur.	Building	Low. Veg.	Tree	Car	Clutter			
Source-Only	72.24	79.15	56.44	35.59	76.85	2.67	76.45	64.83	53.82
RDG	69.85	76.84	59.39	46.48	**84.66**	**24.03**	78.17	72.93	60.21
DRDG	66.86	73.46	58.04	45.18	82.11	15.46	76.16	69.58	56.85
PFST	69.63	87.61	69.56	67.17	80.44	7.90	85.22	72.53	64.20
ST-DASegNet	76.80	84.23	62.41	58.93	81.70	3.80	81.56	70.00	60.74
DAFormer	**80.66**	88.58	63.93	66.58	75.57	15.11	84.73	75.59	65.07
MIC	77.16	**92.54**	64.40	**68.17**	69.53	20.76	85.22	76.51	65.43
Ours	77.66	90.84	**70.42**	64.80	78.04	20.58	**85.76**	**77.62**	**67.06**
Target-Only	87.18	93.80	78.35	80.06	92.17	40.55	91.06	86.73	78.69

2) *Potsdam RGB to Vaihingen Setting:* Table II demonstrates that in UDA tasks across sensors and regions, models trained only on the source domain exhibit very poor segmentation performance for the "Low Vegetation" category. While GAN-based methods also perform inadequately, our method excels in segmenting both the

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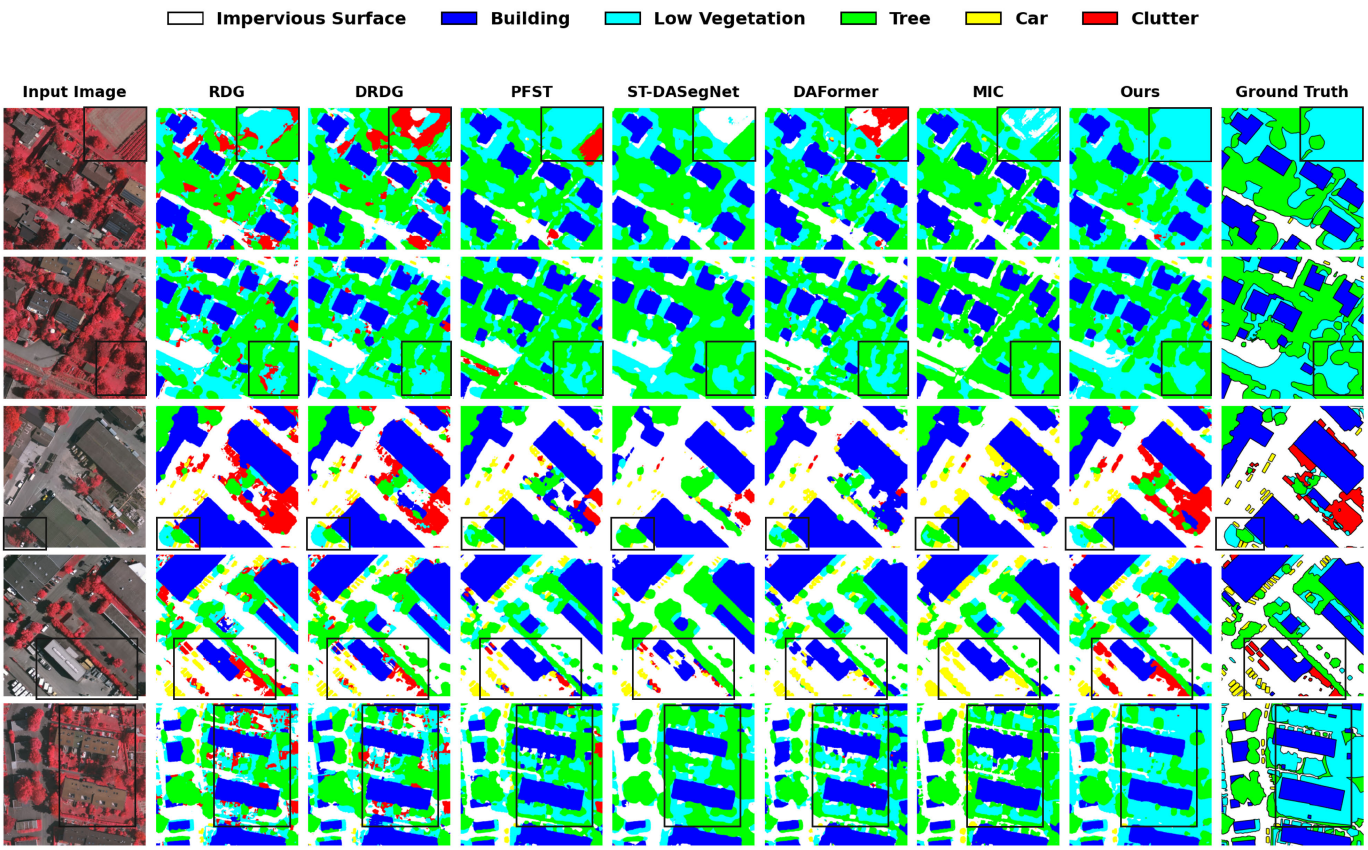


Fig. 4. Visualized semantic segmentation results on the Potsdam RGB to Vaihingen setting.

"Low Vegetation" and "Tree" classes, with a more significant improvement in mIoU compared to other leading UDA methods. This highlights our method's excellent performance not only in scenarios involving the same sensor but also in mitigating the substantial color variation challenges in cross-sensor images through enhanced contextual learning. Additionally, MIC, which also utilizes a masking strategy, shows poor segmentation results for the "Car" and "Clutter" classes, likely due to reduced training for small-scale classes. In contrast, the SRCS proposed in our paper effectively addresses this issue.

3) *Vaihingen to Potsdam IRRG Setting:* Table III reveals that in the UDA task from a small dataset to a large dataset, our method, while still surpassing other UDA methods in various metrics, shows limited improvement. This might be due to the minimal size of the source-domain dataset in this setting, which constrains our method's enhancement of the model's generalization ability. GAN-based methods consistently perform well in segmenting the "Car" across all settings, indicating their stability in small-scale object segmentation, potentially because self-supervised methods often employ ClassMix, leading to poorer performance in segmenting small-scale classes.

To conclude, the results of these three experiments demonstrate that UDA methods significantly advance domain adaptation segmentation tasks, with self-supervised methods currently outperforming GAN-based methods. Our proposed method surpasses the best existing UDA methods in PA, mF1, and mIoU metrics across the three datasets, validating its effectiveness. Furthermore, our method notably improves the segmentation accuracy for confusing classes, showing its proficiency in learning contextual features. Compared to MIC, which also employs a masking strategy, our method achieves better segmentation accuracy for small-scale, few-shot classes, further demonstrating the SRCS method's ability to enhance learning for such classes.

D. Qualitative Results

We present the semantic segmentation results of various UDA methods under three different configurations in Figs. 3–5.

1) *Potsdam IRRG to Vaihingen Setting:* As shown in Fig. 3, in the first and second rows, our method more effectively segments the "Car" and "Clutter" classes. Rows three to five illustrate that other methods frequently misclassify the "Low Vegetation" and "Tree" classes, whereas our method more accurately differentiates between these classes.
2) *Potsdam RGB to Vaihingen Setting:* Fig. 4 demonstrates that in cross-sensor scenarios, the misclassification of the "Low Vegetation" class is significantly more pronounced. In the first row, other UDA methods erroneously segment large portions of the "Low Vegetation"

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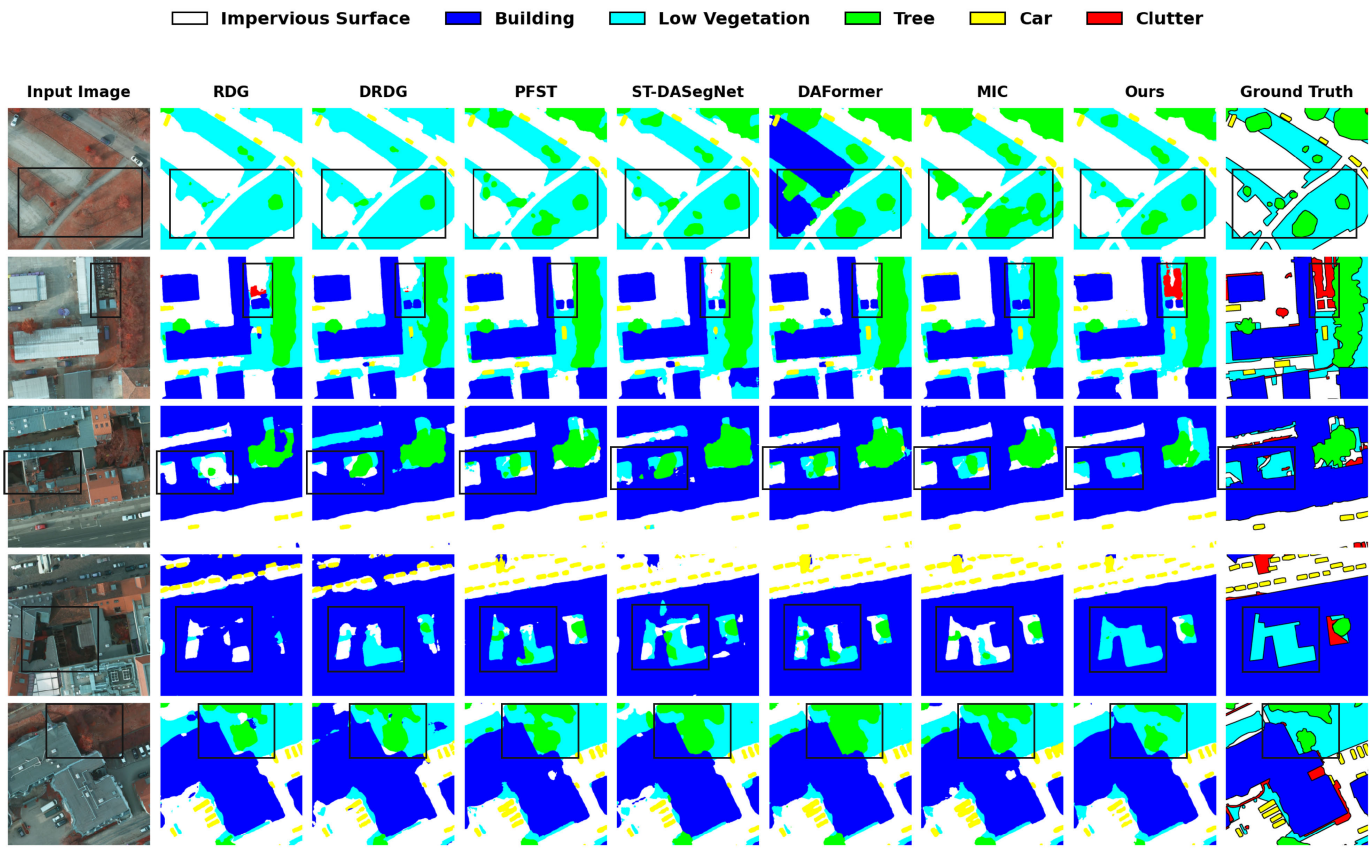


Fig. 5. Visualized semantic segmentation results on the Vaihingen to Potsdam IRRG setting.

areas, while our method achieves precise segmentation. We attribute this to our model's ability to deduce the semantic category of the area based on contextual information, despite the lack of distinct color and texture features. In the third and fourth rows, it is evident that our method markedly outperforms others in recognizing the "Clutter" class.

3) *Vaihingen to Potsdam IRRG Setting:* Fig. 5 illustrates that even with limited source-domain data, our method continues to accurately segment the "Low Vegetation" and "Tree" classes that are often confused in other methods. We hypothesize that this is due to the MECM module increasing the complexity of the learning process, which subsequently enhances the model's robustness.

E. Ablation Study

To assess the effectiveness of the proposed MECM and SRCS, we conducted ablation experiments on the Potsdam IRRG to Vaihingen setting in Table IV. The baseline model used ClassMix instead of MECM and did not incorporate SRCS.

The results indicate that SRCS improves the IoU for multiple classes, especially the small-scale "Car" class. MECM achieves superior segmentation performance compared to ClassMix, particularly for the "Low Vegetation" class, which is prone to confusion. However, the segmentation performance for the "Clutter" class decreased substantially. We attribute this to the fact that the "Clutter" class is a small-scale, few-shot category, and MECM's masking of large image patches may reduce the model's learning opportunities for this class.

With the simultaneous introduction of both SRCS and MECM, the model more effectively learns and segments small-scale, few-shot classes such as "Car" and "Clutter," thereby enhancing overall segmentation accuracy.

Additionally, we compare the training time and memory usage across different experiments. As shown in Table IV, our proposed MECM and SRCS modules enhance performance with minimal impact on training time and memory usage.

Fig. 6 illustrates the confusion matrices of the results for the three models in the transfer from the Potsdam IRRG dataset to the Vaihingen dataset. Fig. 6(a) demonstrates that the baseline method tends to misclassify "Low Vegetation" as "Tree," shows low precision in classifying the "Clutter" class, and frequently misclassifies "Clutter" as "Car." Fig. 6(b) reveals that employing MECM mitigates the confusion between "Low Vegetation" and "Tree" as well as between "Car" and "Clutter." Fig. 6(c) indicates that the combined use of MECM and SRCS further enhances the segmentation accuracy for "Car" and "Clutter," with "Clutter" being less often misclassified as "Car" and exhibiting improved segmentation accuracy.

These experimental findings demonstrate that MECM provides a performance advantage over ClassMix, while SRCS boosts segmentation accuracy for small-scale, few-shot classes.

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TABLE IV

ABLATION STUDY OF THE PROPOSED METHOD ON THE POTSDAM IRRG TO VAIHINGEN SETTING. THE VALUES IN BOLD ARE THE BEST

Method	IoU						PA	mF1	mIoU	Training Time	Memory
	Imp. Sur.	Building	Low. Veg.	Tree	Car	Clutter					
Baseline	82.62	88.66	54.42	74.12	57.52	**50.02**	86.41	79.97	67.89	8.1h	4777MB
With SRCS	82.94	90.88	58.11	75.58	62.08	46.63	87.45	80.95	69.37	8.2h	4777MB
With MECM	**84.97**	92.12	**66.96**	**78.82**	66.45	28.32	88.98	80.02	69.61	8.3h	4789MB
With MECM and SRCS	84.25	**92.98**	66.79	78.80	**69.43**	41.43	**89.35**	**82.76**	**72.28**	8.5h	4789MB

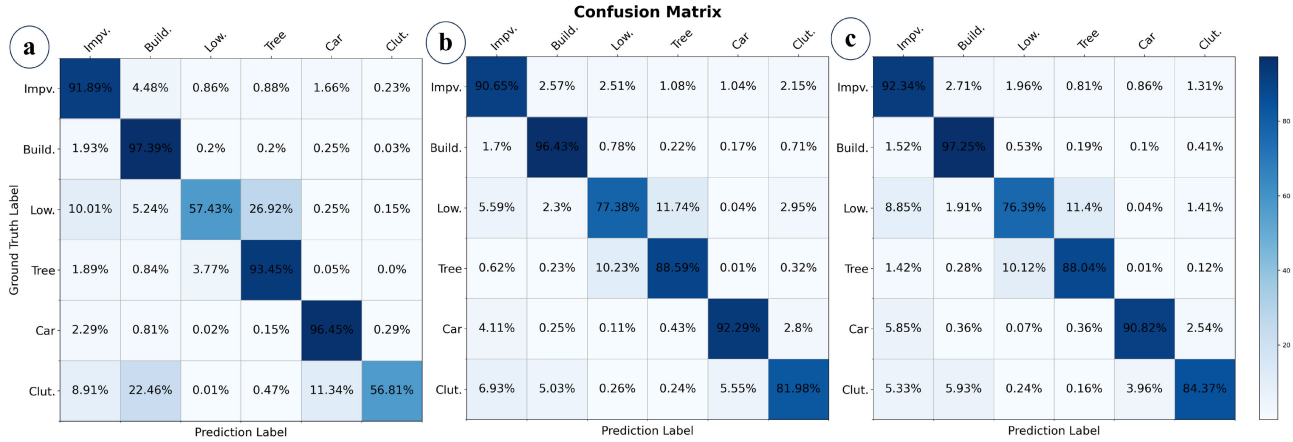


Fig. 6. Confusion matrices of the segmentation results on the Potsdam IRRG to Vaihingen setting. (a) Baseline. (b) With MECM. (c) With MECM and SRCS.

TABLE V

EVALUATION RESULT (%) ON THE POTSDAM IRRG TO VAIHINGEN SETTING WITH DIFFERENT HYPERPARAMETER SETTINGS. THE VALUES IN BOLD ARE THE BEST

Hyperparameters Settings	PA	mF1	mIoU
$p = 0.5$	86.84	78.88	66.92
$p = 0.7$	**88.89**	**80.02**	**69.61**
$p = 0.9$	86.07	78.12	66.74
$t = 0.60, r = 0.05$	88.95	80.71	70.36
$t = 0.75, r = 0.05$	89.04	80.89	70.42
$t = 0.90, r = 0.05$	88.92	80.73	70.15
$t = 0.60, r = 0.15$	88.55	82.72	71.74
$t = 0.75, r = 0.15$	**89.35**	**82.76**	**72.28**
$t = 0.90, r = 0.15$	89.26	82.70	72.19
$t = 0.60, r = 0.25$	88.54	79.57	69.20
$t = 0.75, r = 0.25$	88.65	79.89	69.40
$t = 0.90, r = 0.25$	88.55	79.96	69.40

TABLE VI

COMPARISON RESULTS (%) ON THE POTSDAM IRRG TO VAIHINGEN SETTING OF DIFFERENT SEGMENTATION NETWORKS. THE VALUES IN BOLD ARE THE BEST

Encoder	Decoder	mIoU	Training Time	Memory
MiT-B5	SegFormer	**72.71**	11.5h	6398MB
MiT-B4	SegFormer	72.28	8.5h	4789MB
ResNet-101	SegFormer	68.46	10.4h	5977MB
ResNet-50	SegFormer	67.68	8.3h	5415MB
ResNet-50	DeeplabV3+	66.35	7.5h	5630MB

F. Hyperparameter Settings

To identify the optimal combination of hyperparameters, we conducted experiments in Table V with various hyperparameter settings on the Potsdam IRRG to Vaihingen setting.

First, we use only the MECM method to investigate the impact of different mask ratios p on performance. The experiments show that a setting of $p = 0.7$ achieves better results, with significant improvements over the baseline in three metrics. However, excessively high or low mask ratios reduce the performance of UDA, indicating the sensitivity of the MECM method to the parameter p.

Next, we use the MECM method with $p = 0.7$ to study the performance of different hyperparameter settings for SRCS. The experiments show improvements in all six tests with $r = 0.05$ and $r = 0.15$, demonstrating that increasing the sampling probability for smaller-scale classes effectively alleviates the imbalance issue. However, in the three tests with $r = 0.25$, the model's performance significantly declined compared to the best results, likely due to the over-filtering of samples, which reduced the model's generalization capability. The experiments also show that under the same r setting, the best performance is achieved when $t = 0.75$, indicating that reducing the model's learning on larger scale classes does not diminish generalization capability but rather enhances the overall segmentation performance.

G. Comparison of Network Architectures

In this section, we investigate the impact of different encoders and decoders on UDA performance using different semantic segmentation networks on the Potsdam IRRG to Vaihingen setting.

Experiments in Table VI show that with SegFormer as the decoder, MiT-B4 as the encoder performs significantly better than ResNet-50, which has similar training time and memory usage. Likewise, MiT-B5 outperforms ResNet-101, demonstrating that the newer Transformer-based MiT offers superior feature extraction capabilities and higher UDA efficiency compared to the classic ResNet.

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TABLE VII
COMPARISON OF TRAINING TIME AND MEMORY USAGE FOR DIFFERENT ST METHODS ON THE POTSDAM IRRG TO VAIHINGEN SETTING. THE VALUES IN BOLD ARE THE BEST

Method	mIoU	Training Time	Memory
PFST	68.42	8.4h	8750MB
ST-DASegNet	61.10	11.8h	7542MB
DAFormer	67.89	**8.1h**	**4777MB**
MIC	69.15	12.4h	4788MB
Ours	**72.28**	8.5h	4789MB

Moreover, with ResNet-50 as the encoder, the SegFormer decoder performs better than the classic DeeplabV3+. However, considering that SegFormer requires more training time, the performance gain from the SegFormer decoder is less pronounced than that from a better encoder. This suggests that an encoder with strong feature extraction capabilities has a greater impact on UDA performance.

In summary, different network architectures significantly affect UDA performance. The mIoU of the best network is 6.36% higher than that of the worst, indicating that designing better segmentation networks could be a key focus for future work.

H. Comparison of Training Time and Memory Usage

In this section, we compare the training time and memory usage of different ST methods. Table VII shows that our method achieves the best mIoU results with minimal training time and memory usage, due to our proposed MECM and SRCS modules, which enhance performance with almost no impact on training time and memory consumption.

I. Comparison With Different Segmentation Networks

In this section, we evaluate the generalization of the proposed method by comparing it with the baseline using additional segmentation networks. Specifically, we employ four segmentation networks: DeepLabV2 [51], DeepLabV3+ [52], DANet [53], and SegNeXt [54]. DeepLabV2 and DeepLabV3+ are classic CNN-based semantic segmentation networks that have been widely used in the semantic segmentation of RSIs in recent years. DANet introduces an innovative approach that adaptively integrates local features with global dependencies via the self-attention mechanism. SegNeXt, leveraging convolutional attention, has been experimentally shown to outperform state-of-the-art Transformer-based segmentation networks, establishing itself as one of the leading segmentation networks. DeepLabV2, DeepLabV3+, and DANet are built on the R-50-D8 backbone, while SegNeXt utilizes MSCAN-L. Both our method and the baseline are trained using identical procedures, with the results summarized in Table VIII.

The results indicate that our method consistently outperforms the baseline across all four segmentation networks. Notably, the performance gain of our method over the baseline is more pronounced with more advanced segmentation networks. This is likely because the mask-enhanced strategy benefits from stronger contextual learning capabilities. These findings validate the robustness of our method across various segmentation networks.

TABLE VIII
COMPARISON OF THE PERFORMANCE OF OUR UDA METHOD AND THE BASELINE WITH DIFFERENT SEGMENTATION NETWORKS ON THE POTSDAM IRRG TO VAIHINGEN SETTING. THE VALUES IN BOLD ARE THE BETTER

Network	Method	PA	mF1	mIoU
DeepLabV2	Ours	**84.72**	**79.01**	**66.68**
	Baseline	83.82	76.91	64.24
DeepLabV3+	Ours	**84.42**	**76.52**	**64.75**
	Baseline	83.22	74.86	62.35
DANet	Ours	**86.24**	**80.52**	**68.75**
	Baseline	84.29	78.29	65.71
SegNeXt	Ours	**89.38**	**84.64**	**74.34**
	Baseline	88.14	82.26	71.49

J. Limitations

Since our method reduces misclassification among easily confusable classes and enhances segmentation accuracy for few-shot and small-scale classes, it surpasses previous state-of-the-art techniques in three UDA experiments, achieving the highest performance. However, there is still room for improvement. For example, in the Vaihingen to Potsdam IRRG experiment (Table III), our method's performance remains notably lower compared to models trained with target domain supervision.

In Table I, although our method exceeds the supervised models in the target domain on metrics like mIoU, it falls significantly behind in IoU for the "Low Vegetation" and "Car" classes. In Tables I–III, although our method performs better on the "Car" class segmentation compared to other self-supervised methods, it still lags behind a GAN-based method. This indicates that there is still room for improvement in self-supervised methods for small-scale object segmentation.

In addition, the comparison results of different segmentation networks in Table VI indicate that the segmentation networks significantly affect UDA performance. To further enhance the performance of UDA, our future work will focus on improving semantic segmentation models and exploring the compatibility between different models and UDA methods.

V. CONCLUSION

This article introduces a novel self-supervised framework for mask-enhanced and balanced-sampling UDASS of RSIs, primarily addressing the challenges of class confusion and class imbalance in RSIs. We evaluate the proposed method using three different UDA configurations, and all experimental results demonstrate that our approach significantly improves the segmentation accuracy of frequently confused, few-shot, and small-scale classes, proving its effectiveness. Furthermore, our results show that the proposed method outperforms current state-of-the-art UDA methods in PA, mF1, and mIoU metrics. In the Potsdam RGB to Vaihingen UDA setting, our method's mIoU improves by over 5% compared to leading models. Notably, on the Potsdam IRRG to Vaihingen UDA setting, our method exceeds the performance of models trained with supervision in the target domain on the key metric of mIoU.

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This underscores the significant potential of UDA methods in cross-domain segmentation tasks of RSIs. Moreover, we have experimentally demonstrated the efficiency of our proposed method in terms of training time and memory consumption, with significant improvements over the baseline across various segmentation networks.

However, there remains a need for improvements in segmenting small-scale and few-shot classes. Future work will explore more advanced mask enhancement methods to improve the UDA performance of the model and seek new augmentation methods to further enhance the model's segmentation of small-scale and few-shot classes. Additionally, we will be designing more effective semantic segmentation models to further enhance cross-domain segmentation performance. Due to limited computational resources, the results in this study are based on a single run. With additional resources in future work, we plan to conduct multiple runs and report the average performance along with standard deviations to validate the stability and robustness of our approach.

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